

Exercise 1: Configuring a Basic Spring Application

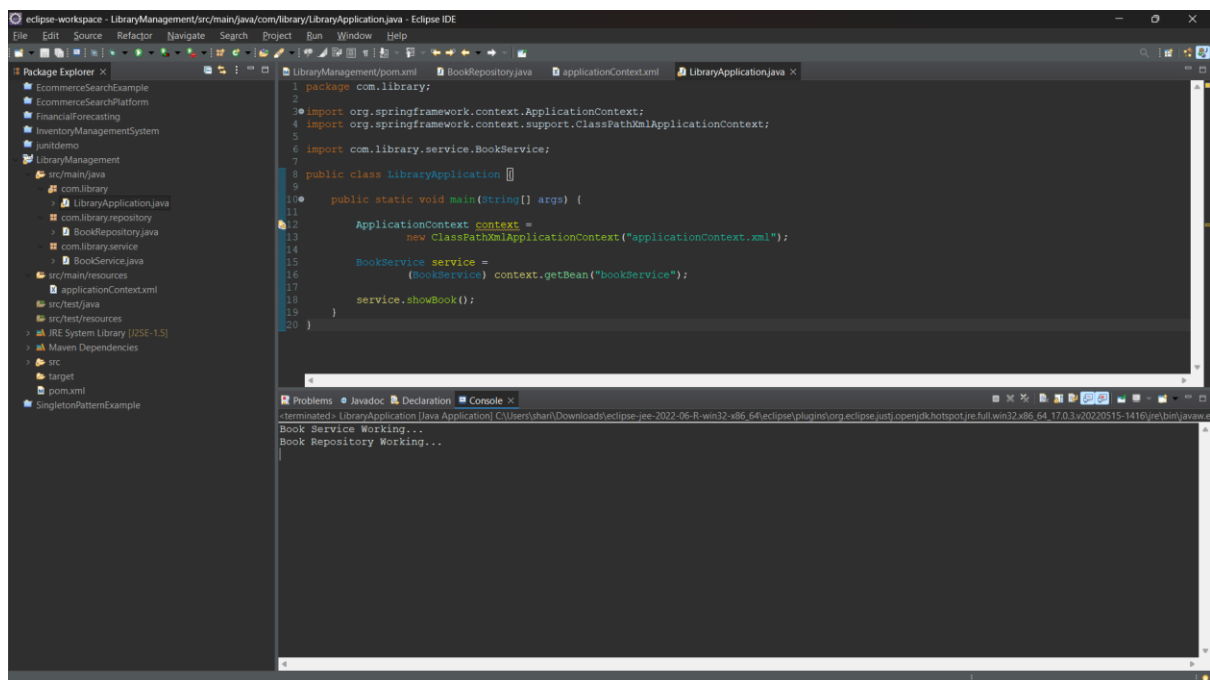
LibraryApplication.java

```
package com.library;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;

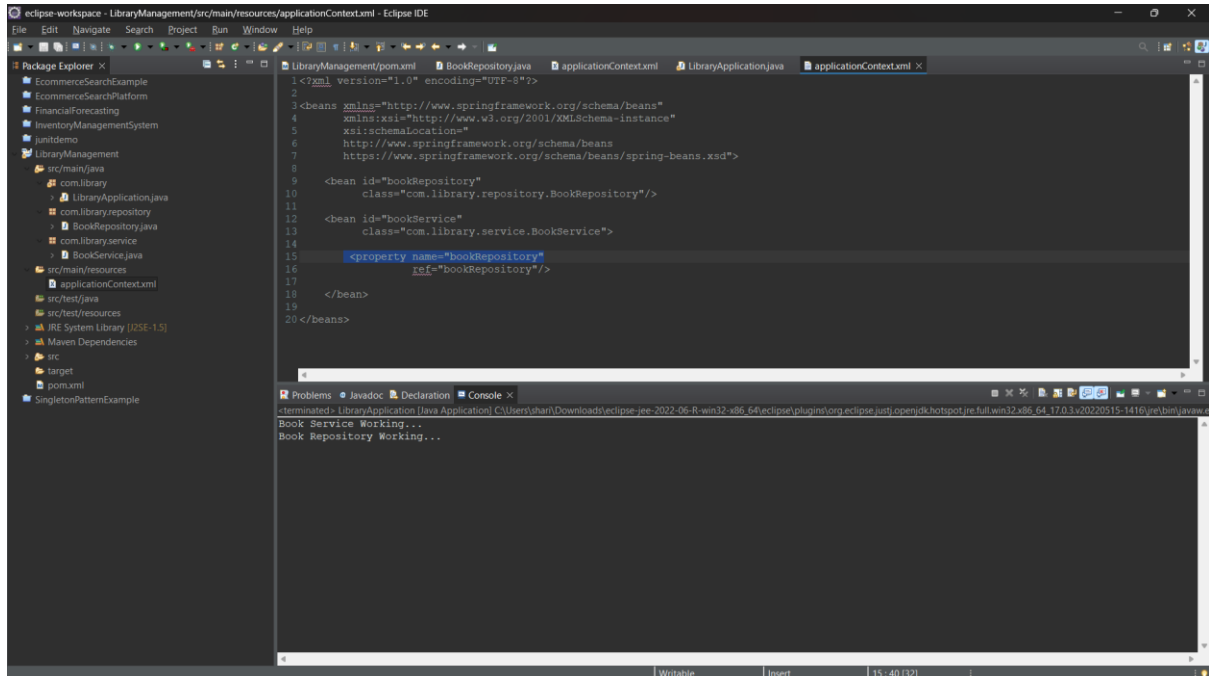
public class LibraryApplication {
    public static void main(String[] args) {
        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");
        BookService service =
            (BookService) context.getBean("bookService");
        service.showBook();
    }
}
```

OUTPUT



Exercise 2: Implementing Dependency Injection

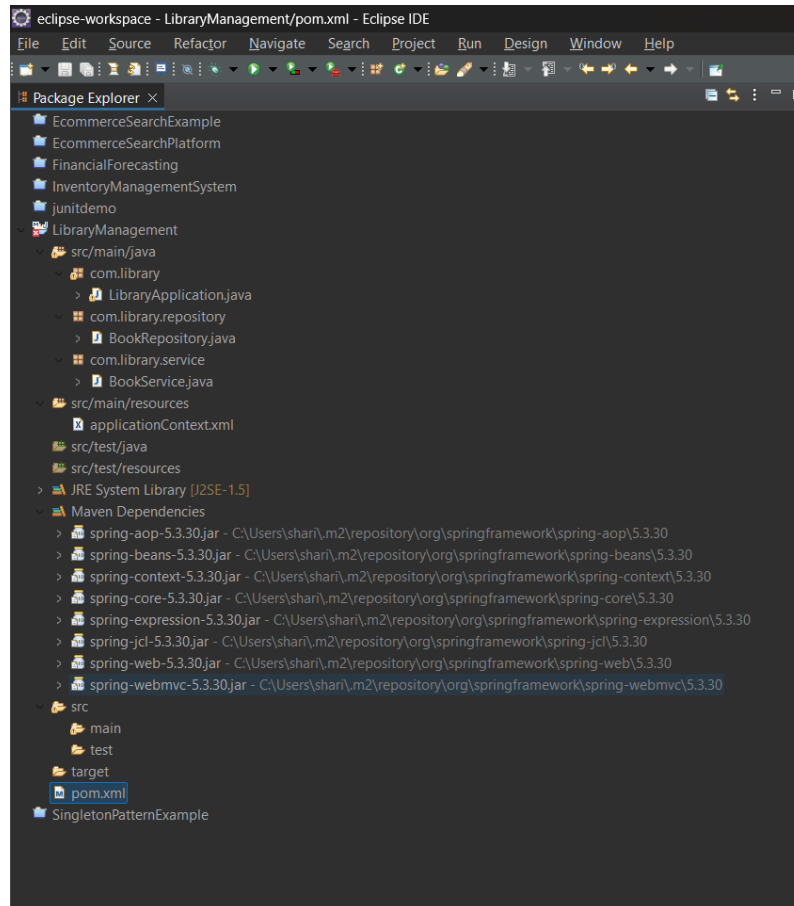
OUTPUT



Successfully implemented Dependency Injection (DI) using Spring Framework. The `BookRepository` bean was injected into the `BookService` bean using Setter Injection through XML configuration. The application executed successfully and demonstrated Spring IoC Container functionality.

Exercise 4: Creating and Configuring a Maven Project

Maven Project Structure



Pom.xml Dependency

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    https://maven.apache.org/xsd/maven-4.0.0.xsd">
```

```
<modelVersion>4.0.0</modelVersion>
```

```
<groupId>com.library</groupId>
```

```
<artifactId>LibraryManagement</artifactId>
```

```
<version>1.0-SNAPSHOT</version>
```

```
<dependencies>
```

```
<dependency>
  <groupId>org.springframework</groupId>
  <artifactId>spring-context</artifactId>
  <version>5.3.30</version>
</dependency>

<dependency>
  <groupId>org.springframework</groupId>
  <artifactId>spring-aop</artifactId>
  <version>5.3.30</version>
</dependency>

<dependency>
  <groupId>org.springframework</groupId>
  <artifactId>spring-webmvc</artifactId>
  <version>5.3.30</version>
</dependency>

</dependencies>

<build>
  <plugins>

    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-compiler-plugin</artifactId>
      <version>3.11.0</version>

      <configuration>
        <source>1.8</source>
        <target>1.8</target>
      </configuration>
    </plugin>

  </plugins>
</build>

</project>
```